

No. of Printed Pages : 4
Roll No.

220733

3rd Sem / Civil
Subject : Surveying - 1

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Which of the following surveys measures the Earth's curvature? (CO1)
- a) Geodetic survey
 - b) Topographic survey
 - c) Hydrographic survey
 - d) Engineering survey
- Q.2 What is the purpose of a bench mark in leveling? (CO2)
- a) Reference point for measurements
 - b) Starting point for survey
 - c) Ending point for survey
 - d) Check point for errors

- Q.3 Which instrument measures vertical angles? (CO3)
- a) Theodolite b) Level
 - c) Compass d) Clinometer
- Q.4 What is the unit of measurement for angular distances in surveying? (CO3)
- a) Degree b) Minute
 - c) Second d) Radian
- Q.5 Which method of surveying uses traverse measurements? (CO4)
- a) Chain surveying b) Compass surveying
 - c) Leveling d) Traverse surveying
- Q.6 What is the purpose of a tie line in surveying? (CO5)
- a) Connects two or more survey lines
 - b) Establishes a reference point
 - c) Measures horizontal distance
 - d) Measures vertical distance

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 The intersection of two or more survey lines is called a _____. (CO5)
- Q.8 The horizontal distance between two stations is called a _____. (CO2)

- Q.9 The difference between magnetic and true north is called _____. (CO2)
- Q.10 The process of adjusting instruments to ensure accuracy is called _____. (CO1)
- Q.11 The type of leveling staff used for precise measurements is called _____. (CO4)
- Q.12 The method of contouring using spot levels is called _____. (CO5)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Explain the concept of arbitrary meridian. (CO2)
- Q.14 Describe the types of chain surveying. (CO1)
- Q.15 What is the purpose of offsetting in chain surveying? (CO1)
- Q.16 Explain the principle of surveying. (CO1)
- Q.17 Describe the method of differential leveling. (CO3)
- Q.18 Explain the principle of radiation in plane table surveying. (CO5)
- Q.19 A surveyor measures the following WCB: (CO2)
- a) $WCB = 57^{\circ} 45'$
 - b) $WCB = 142^{\circ} 30'$
 - c) $WCB = 231^{\circ} 15'$
 - d) $WCB = 322^{\circ} 30'$

Calculate the RB for each and the included angle.

- Q.20 Define contour and write its characteristics. (CO5)
Q.21 Explain the concept of whole circle bearing. (CO2)
Q.22 Write the advantages of Plane table. (CO1)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Define plane table surveying, Explain radiation and traversing method. (CO4)
Q.24 Explain the principles of chain surveying, what operations are carried in chain survey. (CO5)
Q.25 Describe the direct and indirect methods of contouring in detail. (CO6)